## NAPLAN-STYLE PRACTICE -Chapter 1

1 There are 26 845 people at a soccer match. What approximate number could be used to describe the crowd if you round to the leading digit?

2 There are approximately 20 000 000 cattle in Australia. Which of these could *not* be the exact number of cattle?

- 15 674 806
- 19 058 233
- 24 947 321
- 25 004 784

3 After rounding each number to its leading digit, what is the estimate for 33 457 ÷ 167?

- 150
- 200
- 300
- 400

4 What is $4 \times 15 - 5 \times (3^2 + 2)$?

5 What is the missing number in these equivalent fractions? $\frac{40}{72} = \frac{\phantom{0}}{9}$

6 What is $\frac{36}{22}$ written in simplest form?

- $\frac{11}{18}$
- $\frac{9}{11}$
- $1\frac{7}{11}$
- $2\frac{4}{7}$

7 What is the lowest common denominator for $\frac{13}{15} + \frac{3}{5}$?

8 What is $3\frac{3}{5} \times 4\frac{4}{9}$?

- $3\frac{1}{15}$
- $\frac{4}{15}$
- $12\frac{4}{15}$
- 16

9 What is $\frac{5}{21} \div 1\frac{11}{14}$?

- $\frac{125}{294}$
- $\frac{2}{15}$
- $\frac{70}{121}$
- $2\frac{1}{42}$

10 How many decimal places does 125.036 04 have?

11 Which list is in descending order?

- 9.209, 9.82, 8.902, 8.029, 8.29, 8.2
- 9.82, 9.209, 8.902, 8.29, 8.2, 8.029
- 9.209, 8.902, 8.029, 9.82, 8.29, 8.2
- 8.029, 8.2, 8.29, 8.902, 9.209, 9.82

12 What is 39.4951 correct to two decimal places?

- 39.50
- 39.40
- 39.410
- 39.49

13 What is 26.48 + 348.8 + 31.457?

- 405.513
- 406.017
- 406.737
- 91.593

Questions 14–16 refer to this information.

Chocolate bars cost \$2.95. Maria has \$30.

14 How many chocolate bars could Maria buy?

- 8
- 9
- 10
- 11

15 What is the cost of eight chocolate bars?

16 How much change would Maria receive if she bought eight chocolate bars?

17 What is $3.84 \times 6.7$?

- 49.92
- 25.728
- 4.992
- 257.28

18 Which is the best buy: a 2-L bottle of soft drink for \$5.30 or a 1.25-L bottle for \$3.25?

Questions 19–22 refer to these numbers:

$\frac{1}{4}$, 0.763 491..., $\frac{2}{3}$, $\frac{3}{5}$, $\frac{1}{8}$, 8.26, $\frac{9}{11}$.

19 Which can be written as terminating decimals?

20 Which can be written as non-terminating decimals?

21 Which can be expressed as recurring decimals?

22 Which are irrational?

23 What is 8 squared?

16 ▢ 2.8 ▢ $\sqrt{8}$ ▢ 64 ▢

24 What is the square root of 144?

25 What is 6 cubed?

6 ▢ 18 ▢ 36 ▢ 216 ▢

26 What is the cube root of 64?

27 If a square piece of paper has area 100 $cm^2$, how long is the piece of paper?

28 Which of these does *not* simplify to $5^6$?

▢ $\frac{5^9 \times 5^5}{5^8}$ ▢ $\frac{5^4 \times 5^8}{5^5 \times 5^7}$

▢ $(5^2)^3$ ▢ $5^2 \times 5^3 \times 5^1$

Questions 29 and 30 refer to this table.

| $4^2$ | $4^3$ | $4^4$ | $4^5$ | $4^6$ | $4^7$ |
|---|---|---|---|---|---|
| 16 | 64 | 256 | 1024 | 4096 | 16 384 |

29 Find 16 384 ÷ 256 without using a calculator.

30 Find $\frac{1024 \times 64}{4096}$ without using a calculator.

31 Which of these is an irrational number?

$\frac{1}{3}$ ▢ $\sqrt{9}$ ▢ $\sqrt{3}$ ▢ $3^7$ ▢

## ANALYSIS

1 The population figures (rounded to the nearest 100) for each state and territory of Australia at the end of December 2010 are displayed in this table.

| State or territory | Population |
|---|---|
| New South Wales | 7 272 200 |
| Victoria | 5 585 600 |
| Queensland | 4 548 700 |
| Western Australia | 2 317 100 |
| South Australia | 1 650 400 |
| Tasmania | 509 300 |
| Australian Capital Territory | 361 900 |
| Northern Territory | 229 900 |

a Determine the actual difference in population between NSW and Victoria.

b Round each population figure to the leading digit.

Use your answers to part b for these parts.

c Estimate the difference in population between NSW and Victoria. Compare this to part a.

d Estimate the population of Australia.

e Write these populations as fractions in simplest form.

i Victoria as a fraction of NSW

ii Tasmania as a fraction of Queensland

iii NSW as a fraction of WA

iv Queensland as a fraction of Victoria

f Write each answer to part e as a decimal number then round to three decimal places.

g Which of your answers to part f are:

i terminating decimals?

ii non-terminating decimals?

iii recurring decimals?

2 The population of a town is 10 000 and is expected to double every 5 years.

a The town fits within a square of area 9 $km^2$. How long is the square?

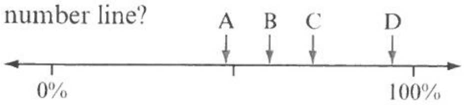
b Use indices to find the population after 30 years.

c It is predicted the town will fit within a square of area 85 $km^2$ after 30 years, how far is it along one side of this square?

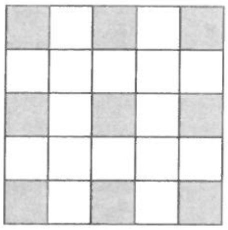
d Classify each of the answers as rational or irrational. Explain your decision.

## NAPLAN-STYLE PRACTICE - chapter 2

1 Which arrow points closest to 70% on this number line?




2 What percentage of the figure has been shaded?



3 What is 42% as a decimal?

0.42 ◯ 4.2 ◯ 42.0 ◯ 4200.0 ◯

4 Which of these is equivalent to $\frac{5}{8}$?

58% ◯ 62.5 ◯ 0.625 ◯ $\frac{5}{800}\%$ ◯

5 Which list of numbers is ordered from smallest to largest?

- ◯ 25%, 7.1, $\frac{1}{2}$, 5.2%
- ◯ 3.6%, 15%, $\frac{7}{8}$, 1.1
- ◯ 128%, $\frac{7}{15}$, 36%, 95
- ◯ $\frac{1}{3}$, $\frac{5}{6}$, 0.905, 15.1%

6 What is $\frac{11}{20}$ as a percentage?

7 Josie scored 35 out of 40 on her Maths test. What percentage is this?

14% ◯ 75% ◯ 87.5% ◯ 114.3% ◯

8 What is 15% of $250?

$265 ◯ $235 ◯ $37.50 ◯ $35 ◯

9 When an item originally priced at $340 has a mark-up of 70%, what is its selling price?

10 Damien notices a sale advertising a discount of 40%. How much would he pay for a coffee machine originally priced at $380?

$532 ◯ $340 ◯ $228 ◯ $152 ◯

11 A car is bought for $22 000 and sold later for $18 500. Which option is correct?

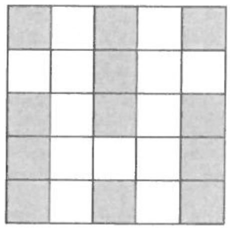
- ◯ A profit of $3500 has been made.
- ◯ A loss of 3500% has been made.
- ◯ A loss of 15.91% has been made.
- ◯ A loss of 20.45% has been made.

Questions 12 and 13 refer to this information.
Ben runs 10 km in 60 minutes. Hayden completes the same run but does it 1.25 times as fast as Ben.

12 Write how much faster Hayden is than Ben, as a percentage.

13 In how many minutes does Hayden complete the run?

14 What is the ratio of non-shaded squares to shaded squares in this diagram?



12:25 ◯ 13:25 ◯ 13:12 ◯ 12:13 ◯

15 Magan's pace length is 65 cm while Michael's is 1.07 m. What is the ratio of Michael's pace length to Magan's?

- ◯ 65:107
- ◯ 65:1.07
- ◯ 1.07:65
- ◯ 107:65

16 Which ratio is *not* equivalent to 3:5:6?

- ◯ 9:15:18
- ◯ 15:25:30
- ◯ 30:15:25
- ◯ 30:50:60

17 Which ratio represents $\frac{3}{4}$ hour to 90 minutes written in simplest form?

1:2 ◯ 5:6 ◯ 45:90 ◯ 2:1 ◯

18 What value of $x$ completes this ratio statement?
$12:x = 132:55$

19 When \$240 is shared in the ratio 1:3, what fraction is the smallest share?

$\frac{1}{3}$ ▢ $\frac{1}{4}$ ▢ $\frac{2}{3}$ ▢ $\frac{3}{4}$ ▢

20 Four students, Mikayla, Morgan, Chelsea and Alexandra, collected money for a charity in the ratio of 5:4:4:2. If the total collected was \$330, how much did Mikayla collect?

21 Tess, Alicia and Alexandra contributed money to a lottery ticket. Tess paid \$2, Alicia paid \$1.50 and Alexandra paid \$2.50. If the ticket won \$1200, how much money would Tess receive if they distributed their winnings in the same ratio as their contribution?

22 Which option best describes an example of a rate?
- ▢ driving 100 km to the beach
- ▢ running as fast as you can to the canteen
- ▢ earning \$15.75 for every hour worked
- ▢ observing the second hand on a clock for one minute

23 A car travels 120 km and uses 10 L of petrol. Which option shows the rate in simplest form?
- ▢ 120 km/L
- ▢ 12 km/L
- ▢ 12 L/km
- ▢ 110 km/L

24 Which is the best value?
- ▢ 8 chocolate bars for \$7.00
- ▢ 3 chocolate bars for \$2.50
- ▢ 5 chocolate bars for \$4.00
- ▢ 13 chocolate bars for \$11.00

## ANALYSIS

A market stall holder bought a variety of coloured 'hoodies' to sell. Each hoodie cost the stall holder \$20 and she intends to sell them to the public at a mark-up of 65%.

a Write the percentage mark-up value as a:
  i fraction in simplest form
  ii decimal.

b Calculate the value of the mark-up on each hoodie.

c What price will the hoodies be?

d If the stall holder sells 35 hoodies on the first day, what is her profit?

To improve the sales of the hoodies for her second day at the market, the stall holder decides to offer a 10% discount on the selling price.

e What is the discount amount for each hoodie?

f What is the new selling price?

g What profit does the stall holder make on the sale of each hoodie?

h The stall holder sells 55 hoodies at the discounted amount. Write the profit from the sale of these as a percentage of the cost she paid for them.

i Of the hoodies sold in part h, 20 were blue, 25 were red and the rest were white. Write the number of blue, red and white hoodies as a ratio in simplest form.

On the first day at the market, the stall holder spent 5 hours selling the hoodies. She spent 8 hours selling them on the second day.

j Write the sales for each day as a rate in simplest form.

k Which day had the best rate of sale?

## NAPLAN-STYLE PRACTICE - chapter 3

1 Which number is the largest?

3 ⬭ −3 ⬭ 2 ⬭ −2 ⬭

2 Which number is smaller than −4.5?

−4.3 ⬭ −4.7 ⬭ $2\frac{1}{2}$ ⬭ $-4\frac{2}{5}$ ⬭

3 What is (+5) + (−9)?

−14 ⬭ −4 ⬭ 4 ⬭ 14 ⬭

4 What is (−6) − (−7)?

−13 ⬭ −1 ⬭ 1 ⬭ 13 ⬭

5 Hayden and Stacy enter a lift at the third level above ground level. They move down five levels. Where does the lift stop?

- ⬭ second level above ground level
- ⬭ second level below ground level
- ⬭ eighth level above ground level
- ⬭ eighth level below ground level

6 Complete this number statement.

17 + ☐ = −25

7 Complete this number statement.

−36 − ☐ = −80

8 A stone falls from a cliff top 50 m above water to the bottom of the sea. The water is 12 m deep. How far does the stone fall?

☐

9 Bread at a temperature of −16°C is taken out of the freezer. After 15 minutes, its temperature rises by 24°C. What is the new temperature of the bread?

−40°C ⬭ −8°C ⬭ 8°C ⬭ 40°C ⬭

10 Complete this number statement.

−4 × ☐ = 12

11 A number is multiplied by −5 and the answer is −30.

The original number is ☐

12 Which pair of numbers has a positive sum and a negative product?

−6, 5 ⬭ −8, −3 ⬭ −4, 7 ⬭ 2, 9 ⬭

13 A ski resort recorded these temperatures: −5°C, 2°C, 4°C, −3°C, −1°C. What is the average of these temperatures?

☐

14 Transactions for Chloe's bank account are shown below.

| Date | Reference | Transaction | Balance |
|---|---|---|---|
| 1 May | – | – | +$124.80 |
| 4 May | Deposit | +$75.00 | |
| 9 May | Music | −$42.75 | |
| 15 May | Shoes | −$149.95 | |
| 26 May | Movies | −$52.20 | |
| 31 May | Deposit | +$150.00 | |

How much is in Chloe's bank account at the end of the month?

☐

15 What is $-\frac{3}{8} \times \frac{4}{9} \times -\frac{2}{7}$?

☐

16 What is $(-2)^5$?

☐

17 Which of these does *not* give a negative result?

$(-5)^6$ ⬭ $(-3)^5$ ⬭ $(-2)^7$ ⬭ $(-4)^3$ ⬭

18 What is $(-3)^{10} \div (-3)^7$?

1 ⬭ −9 ⬭ 27 ⬭ −27 ⬭

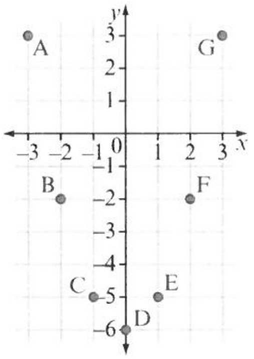
19 A point on the Cartesian plane has the coordinates (−5, 3). What is the $x$-coordinate?

☐

20 What are the coordinates of the origin on the Cartesian plane?

☐

Questions 21–27 refer to this figure.




21 Which point has the coordinates $(2, -2)$?

- A
- B
- F
- G

22 Which point has the coordinates $(-3, 3)$?

23 Write the coordinates of point E.

24 Write the coordinates of point D.

25 Which points are in the third quadrant of the Cartesian plane?

26 Which point has the same $y$-coordinate as point A?

27 Which table of values matches the points plotted on the Cartesian plane?

Table 1

| $x$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |

Table 2

| $x$ | 3 | −2 | −5 | −6 | −5 | −2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |

Table 3

| $x$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | 3 | 2 | 5 | 6 | 5 | 2 | 3 |

Table 4

| $x$ | −3 | −2 | −1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| $y$ | 3 | −2 | −5 | −6 | −5 | −2 | 3 |

## ANALYSIS

The daily minimum and maximum temperatures at a ski resort were recorded over a week.

| | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| Min. temp. (°C) | 0.7 | −2.2 | −0.9 | −3.4 | −5.1 | −1.8 | 0.2 |
| Max. temp. (°C) | 6.5 | 4.1 | 7.8 | 2.5 | −0.6 | 3.9 | 5.3 |

a Which day had:

i the highest temperature?

ii the lowest temperature?

b Calculate the difference between the minimum and maximum temperatures for Monday.

c Calculate the difference between the minimum and maximum temperatures for each other day.

d Which day had the biggest range of temperatures?

e Calculate the average of:

i the minimum temperatures (correct to one decimal place).

ii the maximum temperatures.

f What is the difference between the average minimum and average maximum temperatures?

g Draw a Cartesian plane with the horizontal axis labelled Day (show 1, 2, 3, …, 7 on the scale to represent Monday to Sunday) and the vertical axis Temperature (°C). Think about the scale you will use on the vertical axis to represent all the temperatures listed in the table.

h Write the information for the seven daily minimum temperatures as a set of Cartesian coordinates.

i Plot the seven points with these coordinates on your Cartesian plane. To observe a trend, join the points with a smooth line.

j In a different colour, plot and join points on the same Cartesian plane to show the information for the seven maximum temperatures.

k Describe and interpret the trend you see.

## NAPLAN-STYLE PRACTICE – Chapter 4

1 What is the coefficient of $-8mnp$?

2 Caitlin buys $x$ pens and $y$ pencils. A pen costs \$2 and a pencil costs \$1. Which expression represents the total cost in dollars?

- $x + y$
- $2xy$
- $x + 2y$
- $2x + y$

3 Which of these is equivalent to $5ab$?

- $5 \times a \times b$
- $5 + a + b$
- $5 \times a + b$
- $5 + a \times b$

4 If $a = 4$ and $b = -3$, what is $7a + 2b$?

- $-13$
- 22
- 34
- 97

5 Which of these is equivalent to $-3m^2n$?

- $3 \times m \times n$
- $-3 \times m \times n$
- $3 \times m \times m \times n$
- $-3 \times m \times m \times n$

6 Simplify $5x^2 - 3x + x^2 + 2x - 4$.

7 What is the product of $4ab$ and $3bc$?

- $7abc$
- $12ab^2c$
- $7ab^2c$
- $12abc$

8 Which of these is equivalent to $h^4 \times h \times h^7$?

- $h^3$
- $h^{11}$
- $h^{12}$
- $h^{28}$

9 Which of these shows $\frac{24ef}{20e}$ in simplest form?

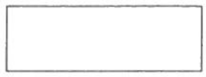
- $\frac{12f}{10}$
- $\frac{6f}{5}$
- $\frac{12ef}{10e}$
- $\frac{6ef}{5e}$

10 Write $b^8 \div b^2$ in simplest form.

11 Simplify $3mn \times 2m^2n \times n$.

12 Simplify $\frac{2x^3 \times 3x^2y^4}{8x^4y^2}$.

13 A rectangle is three times as long as it is wide. If the width is represented by $w$, write an expression for the area of the rectangle in terms of $w$.



$w$

14 Which expression is equivalent to $a(b + c)$?

- $a + (b + c)$
- $a \times b + c$
- $a \times b + a \times c$
- $a \times b \times c$

15 Which expression is equivalent to $5x(x - 2y)$?

- $5x^2 - 2y$
- $5x - 10xy$
- $5x^2 - 5xy$
- $5x^2 - 10xy$

16 If $m = 5$, $n = -2$ and $p = 3$, what is the value of $\frac{2m^2 - 5n}{4mp}$?

17 Which expression becomes $3ab + 6a$ when expanded?

- $3(ab + 2)$
- $3a(b + 2)$
- $3b(a + 2)$
- $3a(b + 6)$

18 Which expression is equivalent to $d(6 - d) + 2d(3d - 1)$?

- $8d$
- $4d + 5d^2$
- $8d + 7d^2$
- $4d - 7d^2$

19 Which expression becomes $4x^2y - 6xy^3$ when expanded?

- $2xy(2x - 3y^2)$
- $2xy(2x - 6y^2)$
- $-xy(4x - 6y^2)$
- $xy(4x + 6y)$

20 Which expression does *not* simplify to $2x - 3y$?

- $\frac{8x^2 - 12xy}{4x}$
- $5x + 3x(y - 1) - 3y(x + 1)$
- $5(x - y) + 3(2y - x)$
- $2(x - 3y) + 3y$

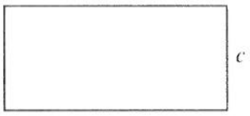
21 Which expression is equivalent to $4p^2 - 10pq$?

- $2p(2p - 5q)$
- $4p(p - 10q)$
- $2(2p - 5pq)$
- $2p^2(2 - 5q)$

22 Which expression is equivalent to $4(m + 3n) - (5m - 2n)$?

- $5n - m$
- $10n - m$
- $12n - m$
- $14n - m$

23 The area of this rectangle is represented by $c^2 + 5c$. What expression can be used to represent the unknown side length?




Questions 24 and 25 refer to a rug with an area given by $(6x^2 + 12x)$ m$^2$.

24 Which are possible measurements for the rug?

- length $(3x + 4)$ m, width $(2x)$ m
- length $(4x + 6)$ m, width $(2x)$ m
- length $(6x + 10)$ m, width $(x)$ m
- length $(2x + 4)$ m, width $(3x)$ m

25 If $x = 2$, what is the length, width and area of the rug?

- 8 m, 6 m, 28 m$^2$
- 12 m, 4 m, 48 m$^2$
- 8 m, 6 m, 48 m$^2$
- 6 m, 4 m, 24 m$^2$

## ANALYSIS

Jared buys a group of friends some boxes of hot chips to share. He wants a way to quickly calculate the cost for each person. The cost depends on whether he buys large or small boxes.

a Jared buys three large boxes of hot chips.
   i What is the total cost?
   ii If the cost is shared between five, how much will each person pay?

b Consider when Jared buys $x$ large boxes of chips.
   i What is the total cost?
   ii If the cost is shared between five, how much will each person pay?

c Write an expression for how much each person pays if $x$ small boxes of chips are bought to be shared between:
   i four people    ii five people.

d Jared considers the cost of buying the same number of large and small boxes of chips.
   i Write an expression, in simplest form, for the total cost of buying $x$ large boxes and $x$ small boxes of chips.
   ii What values could $x$ represent if Jared has \$50 to spend?


e Jared now considers the cost of buying $x$ large boxes of chips and $y$ small boxes of chips. Write an expression for the total cost in:
   i expanded form    ii factorised form.

f For the general case of buying $x$ large boxes and $y$ small boxes of chips, write an expression for the cost per person if the chips are shared between:
   i six people    ii $p$ people.

g Use your expression from part f ii to calculate the cost per person if $x = 3$, $y = 2$ and $p = 6$. Write a sentence to explain this situation.

h For a group of six people, suggest at least three sets of values for $x$ and $y$ so that each person does not spend more than \$5.

i Suggest at least three sets of values for $x$, $y$ and $p$ so that each person does not spend more than \$8.

## NAPLAN-STYLE PRACTICE - chapter 5

1 Which is the solution to $\frac{x}{5} = 5$?

- $x = 25$
- $x = 10$
- $x = 1$
- $x = 0$

2 What is the solution to $3x - 2 = 13$?

3 Using $n$ for the unknown number, which statement matches the equation $2n - 5 = 25$?

- 5 is subtracted from a number and the result is doubled to give an answer of 25.
- A number is doubled then subtracted from 5 to give an answer of 25.
- A number is doubled then 5 is subtracted to give an answer of 25.
- 5 is subtracted from a number to give an answer of 25.

4 Which equation could represent the following statement?

A number is added to 5, then the result is divided by 4 to give an answer of 1.

- $(n + 5) - 4 = 1$
- $4(n + 5) = 1$
- $\frac{n + 4}{5} = 1$
- $\frac{n + 5}{4} = 1$

5 The table below displays the working using the 'guess, check and improve' strategy to solve $3x + 5 = 44$.

| Guess $x$ | LS $3x + 5$ | RS 44 | Check Is LS = RS? |
|---|---|---|---|
| 10 | 35 | 44 | no |
| 15 | 50 | 44 | no |
| 12 | 41 | 44 | no |
| 13 | 44 | 44 | yes |

Which of the following statements is *not* correct?

- The value of $x$ for the solution is between 10 and 15.
- The value of $x$ for the solution must be greater than 15.
- The solution is $x = 13$.
- $x = 12$ is not the solution to the equation.

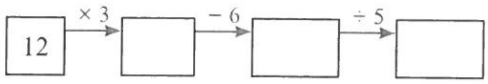
Questions 6 and 7 relate to this information. Entry to a fun park costs \$15 plus \$3 for every ride. Mikayla takes \$75 to the park.

6 If $r$ represents the number of rides, which equation matches this information?

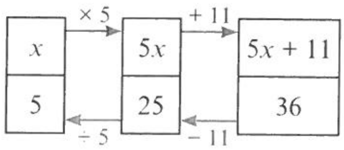
- $r + 15 = 75$
- $15r + 3 = 75$
- $3r + 15 = 75$
- $75 + 3r = 15$

7 How many rides can Mikayla have?

8 What number will appear in the final box of this flowchart?

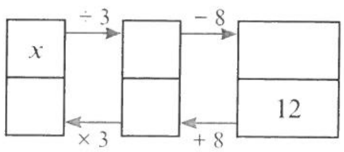



9 This flowchart shows $5x + 11 = 36$.




What is the solution to the equation?

Questions 10 and 11 relate to this flowchart.




10 Which equation does it represent?

- $\frac{x - 8}{12} = 12$
- $3x + 8 = 12$
- $\frac{x}{3} - 8 = 12$
- $3(x + 8) = 12$

11 Which of these is the solution?

- $x = 4$
- $x = 12$
- $x = 20$
- $x = 60$

12 Which equation is equivalent to $2x + 3 = 15$?

- $x = 9$
- $x = 10$
- $2x = 12$
- $2x = 18$

13 Which of these is the equivalent equation formed when 12 is subtracted from both sides of $4x = 32$?

- $4x - 12 = 20$
- $-8x = 20$
- $4x - 12 = 32$
- $12 - 4x = 20$

14 Which equation is *not* equivalent to $x + 4 = 15$?

- $x + 9 = 20$
- $x = 11$
- $2x + 8 = 30$
- $x + 8 = 11$

15 What is the solution to $12 - 4x = 24$?

- $x = 9$
- $x = 3$
- $x = -3$
- $x = -9$

16 Which operation should be performed first on both sides of $5x + 6 = 3x - 10$ to remove one of the pronumeral terms?

- $+ 3x$
- $- 3x$
- $+ 10$
- $+ 5x$

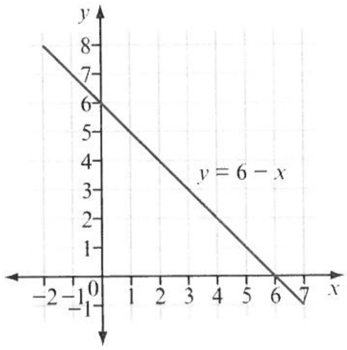
17 What is the solution to $6x - 8 = 4x - 12$?

18 Which linear equation matches the values in the table?

| $x$ | −2 | −1 | 0 | 1 | 2 |
|---|---|---|---|---|---|
| $y$ | −4 | −1 | 2 | 5 | 8 |

- $y = x + 2$
- $y = 3x + 2$
- $y = 2x + 4$
- $y = x - 2$

Questions 19 and 20 refer to the graph of $y = 6 - x$ shown at right.




19 What is the $x$ value when $y$ is 7?

20 Using the graph, what is the solution to $6 - x = 4$?

- $x = 2$
- $x = 3$
- $x = 4$
- $x = 6$

## ANALYSIS

A party venue costs \$150 to hire, plus \$20 per person for games and food.

a If $x$ people attend the party, write an expression for the cost.

b Kacie's mother gives her \$450 for venue hire, games and food. Use part a to write an equation.

c Solve the equation in part b in two different ways to find $x$. What does this value represent?

d Kacie would like to invite 24 friends. Calculate the total cost.

e Will Kacie be able to afford this many guests at her party? Justify your answer.

Kacie hears of another party venue that offers a similar service. Their costs include a hire fee of \$200 plus \$10 per person for games and food.

f Write a new expression for the cost of $x$ people attending the party.

g Form an equation and work out how many people Kacie can have at the new venue.

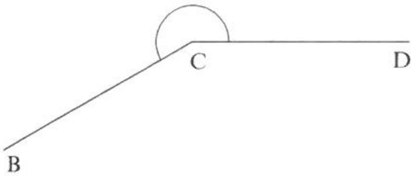
h Can Kacie afford all her guests at the new venue?

i Using the cost expressions for the two venues, how many people could attend the party for the same cost at either venue? (Hint: form an equation from the two expressions and solve for $x$.)

j Use part i to work out the hire cost.

## NAPLAN-STYLE PRACTICE - chapter 6

Questions 1–3 refer to this angle.




1 What is one name for the angle?

- ∠CDB
- ∠DBC
- ∠CBD
- ∠DCB

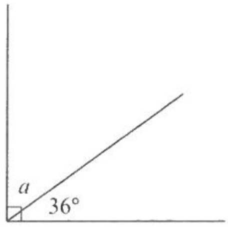
2 What type of angle is shown?

- an acute angle
- a reflex angle
- a right angle
- an obtuse angle

3 What is the size of the angle?

- 330°
- 230°
- 210°
- 150°

Questions 4 and 5 refer to this figure.

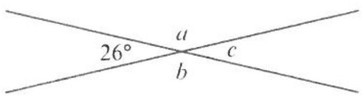



4 What is the value of $a$?

5 What name could you give these angles?

- vertically opposite
- supplementary
- alternate
- complementary

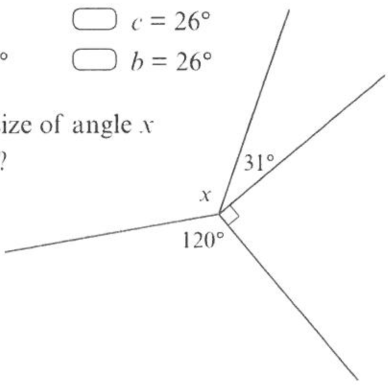
6 A number of angles are created when two lines cross.



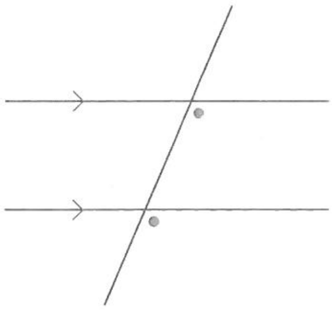

Which of these is *not* true?

- $a = b$
- $c = 26°$
- $a = 154°$
- $b = 26°$

7 What is the size of angle $x$ in this figure?

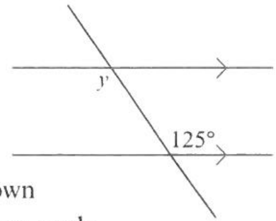



8 Which statement best describes this figure?



- the angles are corresponding
- the angles are co-interior
- the angles are alternate
- the angles are supplementary

9 What is the size of angle $y$ in this figure?



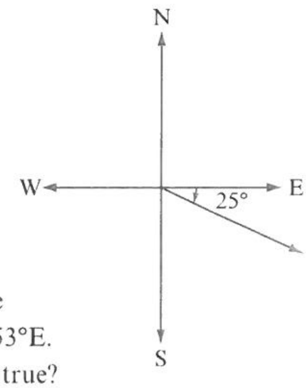

10 A triangle has two known side lengths and a known angle between them. Which description matches this statement?

- SSS
- ASA
- SAS
- AAS

11 What is N52°W as a true bearing?

- 308°T
- 352°T
- 052°T
- 248°T

12 What bearing is shown here?




- E25°S
- S65°E
- S25°E
- 025°T

13 Brisbane is near the coordinates 27°S 153°E. Which statement is true?

- the line of longitude is 153°E
- the line of latitude is 153°E
- the equator runs through Brisbane
- the line of longitude is 27°S

14 What are the coordinates for a location with a latitude of 40° north of the equator and a longitude of 14°E?

- 40°N 14°E
- 40°E 14°N
- 40°S 14°E
- 40°N 40°E

15 How many minutes are there between 11.48 am and 3.15 pm?

Questions 16–18 refer to this information.

Mexico City is 6 hours behind UTC. Darwin is 9.5 hours ahead of UTC.

16 How many hours difference is there between Mexico City and Darwin?

17 If it is 3 pm Saturday in Mexico City, what is the time and day in Darwin?

- 12.30 am Saturday
- 11.30 pm Friday
- 11.30 pm Saturday
- 6.30 am Sunday

18 If it is 3 pm Saturday in Darwin, what is the time and day in Mexico City?

- 12.30 am Saturday
- 11.30 pm Friday
- 11.30 pm Saturday
- 6.30 am Sunday

19 Auckland (New Zealand) is 2 hours ahead of Sydney (Australia). A plane leaves Sydney at 1 pm and takes 3 hours and 45 minutes to reach Auckland. What is the time in Auckland when the plane lands?

- 4.45 pm
- 2.45 pm
- 6.45 pm
- 5.45 pm

## ANALYSIS

Inspector Claude of Interpol is on the trail of the international art thief known as the Hyena. His pursuit leads him from London to Rome, from Rome to Prague and from Prague back to London. Use the map on page 321 to answer these questions.

a What are the three cities' latitudes and longitudes?

b Using the lines of longitude as north–south lines, what is the true bearing for each of these flights?

  i London to Rome

  ii Rome to Prague

  iii Prague to London

c Inspector Claude's flight from London to Rome takes 2 hours and 25 minutes. If he leaves London at 9 am, what time will it be when he lands in Rome?

d Before Interpol will pay for his flights, Inspector Claude needs to submit a map of where he travelled. Use a ruler and protractor to draw the triangle LRP (London, Rome, Prague). Decide if you will use the SSS, ASA or SAS method. Sketch the European coast line around your triangle and remember to make sure all the bearings are accurate.

e The Inspector spends a few hours in each city before finding the clue that reveals the Hyena's next location. Below are the details of how long he spends in each city and the duration of each flight.

| | |
|---|---|
| London to Rome | 2 h 25 min |
| Investigating Rome | 2 h 17 min |
| Rome to Prague | 1 h 50 min |
| Investigating Prague | 1 h 35 min |
| Prague to London | 1 h 45 min |

How much time (in hours and minutes) has passed before Inspector Claude returns to London?

f In Prague, the Inspector learns that the Hyena plans to bring a package to London airport at 7 pm. If Inspector Claude first left London at 9 am, will he make it back there in time to catch the Hyena?

## NAPLAN-STYLE PRACTICE - chapter 7

Questions 1 and 2 refer to the following statement. Devon measured the angles of a triangle and found that angles *b* and *c* were twice the size of angle *a*.

1 What type of triangle is this?

- ☐ right-angled triangle
- ☐ isosceles triangle
- ☐ equilateral triangle
- ☐ scalene triangle

2 What is the size of angle *a*?

| 30° | 36° | 60° | 72° |
|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ |

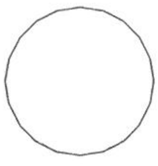
3 Alexandria found that one pair of sides of a quadrilateral were parallel. What type of quadrilateral can it *not* be?

- ☐ trapezium
- ☐ kite
- ☐ rhombus
- ☐ rectangle

4 A parallelogram has one angle of 23°. What is the size of the other different angle?

☐

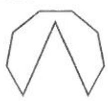
5 An icosagon is a polygon with 20 sides. Using the triangle method, what is the internal angle sum of an icosagon?

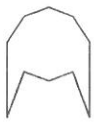


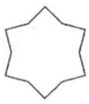
| 20° | 162° | 360° | 3240° |
|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ |

6 Which object is a concave decagon?

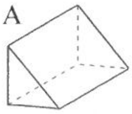
☐ A

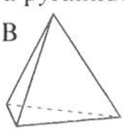

☐ B


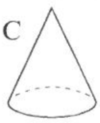
☐ C


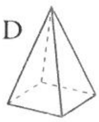
☐ D


7 Which of the following is *not* a pyramid?

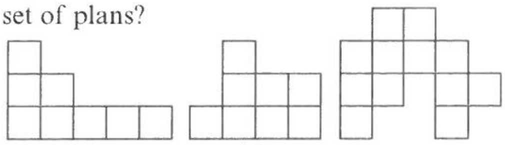
☐ A


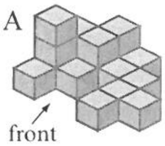
☐ B


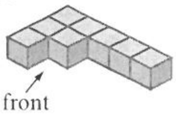
☐ C


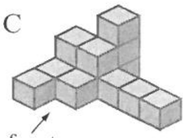
☐ D


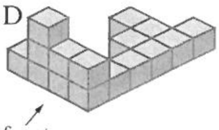
8 Which object matches this set of plans?



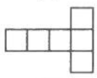

☐ A


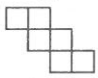

☐ B


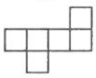

☐ C


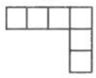

☐ D



9 Which of these nets does *not* fold up to form a cube?

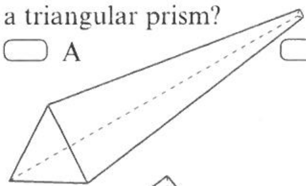
A

☐

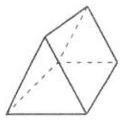
B

☐

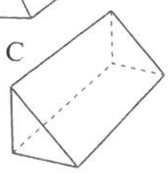
C

☐

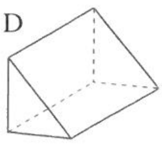
D

☐

10 Which drawing is *not* a perspective drawing of a triangular prism?

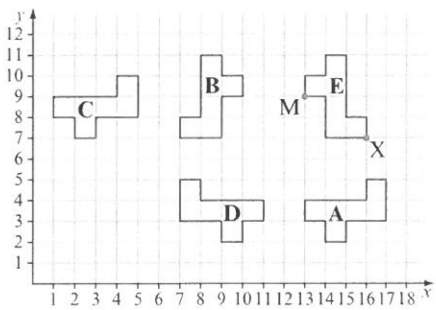
☐ A


☐ B


☐ C


☐ D


Questions 11–13 refer to the following figure.




11 Which shape is a rotation of B?

| A | C | D | E |
|---|---|---|---|
| ☐ | ☐ | ☐ | ☐ |

12 What is the transformation from C to A?
- reflection in a diagonal mirror line
- translation of 12 left, 5 up
- translation of 5 left, 12 down
- translation of 12 right, 5 down

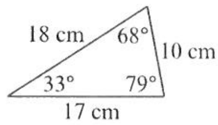
13 Shape E was reflected in a vertical mirror line at point M and then translated 2 units down and 3 units right. What are the coordinates of corner X after these transformations?
- (13, 5)
- (10, 5)
- (16, 7)
- (7, 16)

14 Which sequence of transformations does *not* produce congruent figures?
- reflection, translation, reflection
- rotation, translation, dilation
- translation, rotation, translation
- reflection, rotation, translation

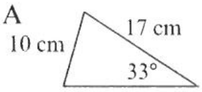
15 Which of these conditions cannot be used to determine congruence?

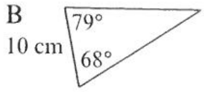
SSA SAS SSS ASA

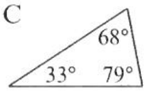
Questions 16 and 17 refer to this triangle.

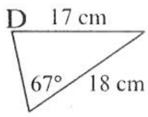



16 Which of these triangles is congruent to the original triangle?

A



B


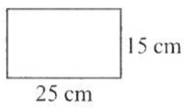

C



D



17 Which condition for congruence did you use to answer question 16?

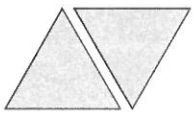
SSS ASA SAS RHS

18 The rectangle shown was dilated so that the shortest side measured 5 cm. What is the scale factor for this dilation?




## ANALYSIS

A chocolate company is making a new product and needs your help with the design. Each chocolate is an equilateral triangle with side length 2 cm and they are to be stacked in a prism, as shown in this front view.



a In what ways can you describe one triangle as a transformation of the other?

b What shape is the front view of the prism? Draw this shape and label it with its angles and side lengths.

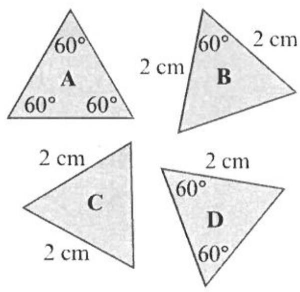
c Name the prism of this shape.

d Use isometric dot paper to draw this prism.

e Draw the following representations of this package.

i a net
ii a perspective drawing
iii a set of plans

The company produces the chocolates shown here.




f Which chocolate(s) can you be certain are the right dimensions? Explain how you know.

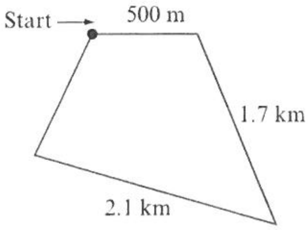
g The company also wants to produce a jumbo size in which the dimensions are doubled. State the scale factor for this dilation and draw the new chocolates and packaging, labelling the angles and side lengths.

## NAPLAN-STYLE PRACTICE - chapter 8

1 Juan is building a model aeroplane. He needs a plank of wood that is 1150 mm long. If he buys a plank of wood 1.5 m long, how many centimetres will he have to cut off?

1.15 ◯ 350 ◯ 45 ◯ 35 ◯

2 The total length of a cross-country course is 5 km. What is the length of the last leg?




◯ 700 m
◯ 4.3 km
◯ 9.3 km
◯ 430 m

3 A vegetable garden is in the shape of a rectangle. The total length of fencing is 3.4 m and the garden is 1 m long. How wide is the garden?

☐

4 A circular sushi hand roll measures 4.5 cm across its diameter. What length of seaweed wraps around the circumference?

☐

5 A circular hat measures 65 cm around the brim. How wide is it across its diameter?

◯ 204.2 m ◯ 20.69 km
◯ 13 273 cm ◯ 10.35 cm

Questions 6 and 7 refer to a car wheel with a radius of 20 cm.

6 What is the length of one revolution of the wheel?

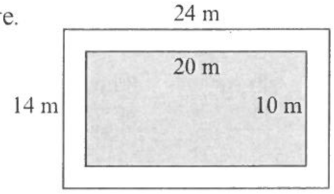
◯ 20 cm ◯ 62.83 cm
◯ 125.66 cm ◯ 1256.64 cm

7 By how much would this radius need to increase if the wheel had to cover 1.5 m in one revolution?

◯ 24.34 cm ◯ 12.17 cm
◯ 7.75 cm ◯ 3.87 cm

Questions 8 and 9 refer to the following information and figure.

A pool has concrete pavers around its edges, as shown in the figure.



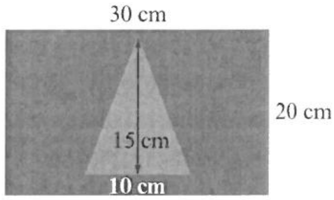

8 What is the area of the swimming pool?

☐

9 What is the area covered by the concrete pavers?

◯ 336 m$^2$ ◯ 136 m$^2$
◯ 76 cm$^2$ ◯ 38 m$^2$

10 Derek designed this flag for his sports team. What is the area of the blue section?




◯ 75 cm$^2$ ◯ 450 cm$^2$
◯ 525 cm$^2$ ◯ 600 cm$^2$

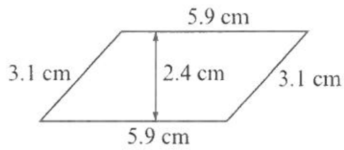
11 A triangle has an area of 40 cm$^2$. If it is 8 cm high, how long is its base?

☐

12 Brittany has a kite that is 75 cm long and 40 cm wide. What is the area of this kite?

☐

13 What is the area of this shape?



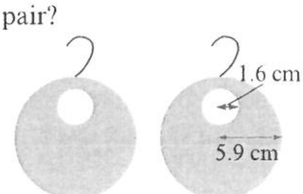

☐

14 A Frisbee has a diameter of 25 cm. What is its area?

☐

15 A pair of earrings is to be made according to the blueprints shown. How much metal is needed to make the pair?

◯ 218.72 cm$^2$
◯ 202.63 cm$^2$
◯ 109.36 cm$^2$
◯ 101.32 cm$^2$



16 What is the diameter of the largest plate that could be made with 200 $cm^2$ of porcelain (to the nearest centimetre)?

4 cm ▢ 8 cm ▢ 16 cm ▢ 64 cm ▢

Questions 17–19 refer to the following information.

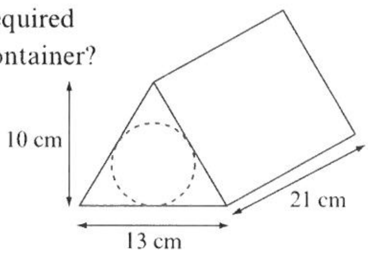
Tennis balls are placed into a container that is a rectangular prism with the dimensions 7 cm by 7 cm by 21 cm.

17 How much cardboard is required to make this container?

▢

18 The tennis balls are instead placed inside a container in the shape of an equilateral triangular prism, as shown. How much cardboard is required to make this container?

▢ 44 $cm^2$
▢ 949 $cm^2$
▢ 1079 $cm^2$
▢ 2730 $cm^2$




19 What is the difference in volume between these containers?

▢ 263 $cm^3$ ▢ 336 $cm^3$
▢ 1029 $cm^3$ ▢ 1701 $cm^3$

Questions 20–22 refer to the following information.

Katie has a vase in the shape of a rectangular prism. Its base measures 5 cm by 5 cm and it is 20 cm high.

20 What is the volume of the vase?

▢

21 What amount of water will fit inside the vase?

5 L ▢ 50 L ▢ 500 mL ▢ 25 mL ▢

22 How many vases would be required to hold 1 $m^3$ of water?

2 ▢ 20 ▢ 200 ▢ 2000 ▢

23 A sheet of glass has an area of 3500 $mm^2$. What is this measurement in $cm^2$?

▢

## ANALYSIS

A friend wants to have a ball pit at their birthday party. They have set aside a rectangular space in their backyard that is 3 m long and 2 m wide.

a What area would this cover?

b How much fencing would be needed to go around the ball pit?

Your friend can only buy the wood that they need for the fencing in planks that are 50 cm wide. The walls of the pit need to stand 1 m tall.

c What length of wood is needed?

d Your friend wants to paint the wooden fencing inside and out with two coats of paint. What is the total surface area to be painted?

They also want to paint a diamond on the front fence of the ball pit. From point to point (that is, its diagonals) it measures 55 cm vertically and 30 cm horizontally.

e What is the area of this diamond?

Around the diamond will be six circles representing the coloured balls in the ball pit. Each ball has a diameter of 6.6 cm.

f What is the circumference of each circle?

g Find the total area to be painted for the six balls.

h What will be the volume within the pit?

i If each ball to go into the pit has a volume of 150 $cm^3$, how many could theoretically fit into the pit? Be careful with unit conversions!

j Your friend finds that only about 25 000 balls fit into the pit. What explanation can you offer for this difference?

## NAPLAN-STYLE PRACTICE -chapter 9

1 What sampling method is surveying every 10th person who walks into a store?

- systematic sampling
- random sampling
- stratified sampling
- biased sampling

2 Which of these is an example of a census?

- Asking 1 000 000 Australians if they want to change the Australian flag.
- Asking everyone in your class if Year 8 students prefer Coke or Pepsi.
- Asking every person in Victoria if people in Hobart like the prime minister.
- Asking everyone in your class if the class prefer cats or dogs.

Questions 3–5 refer to this data set.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 12 | 35 | 21 | 19 | 18 | 18 | 39 | 11 |
| 29 | 31 | 28 | 27 | 14 | 31 | 30 | 20 |
| 19 | 15 | 21 | 22 | 28 | 31 | 37 | 18 |

3 What is the best size for class intervals?

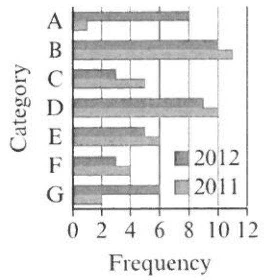
- 30
- 10
- 5
- 1

4 What type of data is this?

- numerical, discrete
- numerical, continuous
- categorical, nominal
- categorical, ordinal

5 If the first class interval is 10–14, what is the most popular class?

Questions 6 and 7 refer to this graph.




6 What is the least popular category overall?

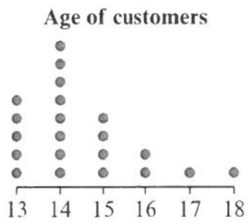
- A
- C
- F
- G

7 How many people were surveyed in 2011?

8 How is the relationship between height and weight best represented?

- column graph
- pie graph
- line graph
- scatterplot

9 Which sentence best describes this dot plot?



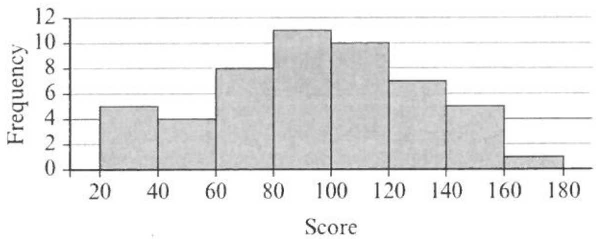

- Customers are likely to be teenagers.
- The least common age is 18.
- The youngest customer is 13.
- Customers are most likely to be in their early teens (13–15 years).

10 Which key shows the most common value in this stem-and-leaf plot is 1200?

| Stem | Leaf |
|---|---|
| 0 | 7 8 9 9 |
| 1 | 1 1 2 2 2 2 2 8 9 |
| 2 | 0 1 3 5 6 |
| 3 | 0 7 |
| 4 | |
| 5 | 1 |

- 1 | 1 = 11
- 1 | 1 = 1100
- 11 | 00 = 1100
- 110 | 0 = 1100

Questions 11 and 12 refer to this graph.




11 What is the modal class?

- 20–40
- 20–180
- 80–100
- 160–180

12 The shape of the histogram can best be described as:

- roughly symmetrical
- roughly symmetrical with an outlier
- skewed
- skewed with an outlier

Questions 13–15 refer to the data in this table.

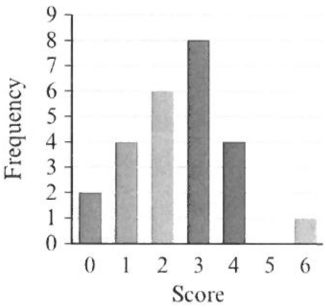
| Score | Frequency |
| --- | --- |
| 0 | 2 |
| 1 | 7 |
| 2 | 8 |
| 3 | 2 |
| 4 | 1 |

13 What is the median?

14 What is the mean?

15 What is the range?

16 What is the mean, median, mode and range for this graph?




- 3, 3, 8, 8
- 2.48, 3, 3, 6
- 3.57, 3, 3, 8
- 2.5, 3, 3, 6

17 There is an issue important to everybody in your postcode. What would be the best sample to survey?

- asking 50 people you know
- asking 50 random people at the shopping centre
- asking 50 random people from the electoral roll
- asking 200 people in the state

Questions 18 and 19 refer to this situation.

Bulaale wants to find the most popular TV show watched by the Year 8s at his school. He surveys two classes of Year 8 students (40 students out of a total of 200) and finds that, out of the 10 TV shows mentioned, the most popular one is *The Simpsons*.

18 Which summary statistic is Bulaale using when he names *The Simpsons*?

19 What might prevent Bulaale from using this sample to predict the most popular TV show for students in Year 8?

- The sample was not big enough.
- The surveying technique was biased.
- *The Simpsons* is probably not the most popular TV show.
- There are more than 10 shows on TV.

## ANALYSIS

Drew wanted to find the average age of people who went to the local beach. One Saturday he went to the beach and asked the age of every fifth person who walked past. The results are shown.

| | | | | | | | | | |
| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 14 | 17 | 18 | 23 | 21 | 19 | 23 | 45 | 23 | 46 |
| 61 | 15 | 16 | 17 | 39 | 32 | 9 | 19 | 27 | 26 |
| 31 | 5 | 21 | 19 | 18 | 16 | 15 | 26 | 22 | 59 |
| 72 | 13 | 49 | 43 | 16 | 15 | 19 | 23 | 22 | 21 |
| 63 | 56 | 8 | 24 | 35 | 16 | 19 | 25 | | |

a What type of data is this?

b Describe the sampling method. Is Drew taking a census or a sample?

c Put the data into a frequency table with appropriate class intervals.

d Draw an appropriate graph of the data.

e Create a stem-and-leaf plot of the data.

f Calculate summary statistics on the data.

g What limitations does this sample have if you want to use it to predict the characteristics of the population?

h Make a prediction about what you would expect to see in the population.

## NAPLAN-STYLE PRACTICE -Chapter 10

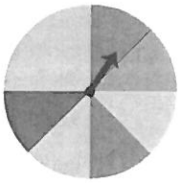
1 Alicia has 10 highlighters in her pencil case, four yellow, two green, one blue and the rest are pink. What is the sample space of the pencil case?

- ▢ {yellow, green, blue}
- ▢ 10
- ▢ {4 yellow, 2 green, 1 blue, 3 pink}
- ▢ {yellow, green, blue, pink}

2 Peter flips two coins and records the outcome. What is the probability of flipping at least one head?

▢

Questions 3–5 refer to this spinner.



3 What is the probability of spinning yellow?

▢ $\frac{1}{4}$ ▢ $\frac{1}{2}$ ▢ $\frac{3}{7}$ ▢ $\frac{3}{8}$

4 What is the probability of spinning green or red?

▢ $\frac{1}{4}$ ▢ $\frac{1}{2}$ ▢ $\frac{2}{7}$ ▢ $\frac{1}{8}$

5 What is the complementary event to spinning green?

▢ red ▢ $\frac{7}{8}$

▢ yellow, red or blue ▢ $\frac{1}{8}$

6 An event, A, has a probability of 0.2. What is the probability of its complementary event, A′?

▢

Questions 7–9 refer to this two-way table.

| | Year 7 | Year 8 | Total |
|---|---|---|---|
| Female | 3 | 8 | 11 |
| Male | 2 | 7 | 9 |
| Total | 5 | 15 | 20 |

7 What is the probability that a person chosen at random from the group is a female in Year 8?

▢

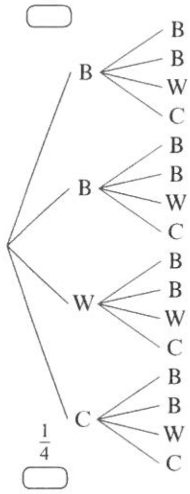
8 What is the probability that a person chosen at random from the group is male?

▢ $\frac{1}{4}$ ▢ $\frac{9}{20}$ ▢ $\frac{1}{10}$ ▢ $\frac{11}{20}$

9 What is the probability that a person chosen at random from the group is in Year 8, given that they are male?

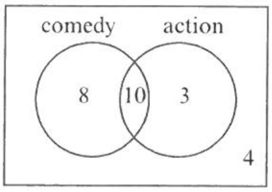
▢ $\frac{7}{9}$ ▢ $\frac{7}{20}$ ▢ $\frac{9}{20}$ ▢ $\frac{7}{15}$

10 A drawer contains loose socks that are black, white or coloured. There are twice as many black socks as there are white or coloured socks. Two socks are selected at random from the drawer, as shown in this tree diagram. What is the probability that the two socks drawn will match?




▢ $\frac{1}{3}$ ▢ $\frac{1}{16}$ ▢ $\frac{3}{8}$ ▢ $\frac{1}{4}$

Questions 11 and 12 refer to this Venn diagram.

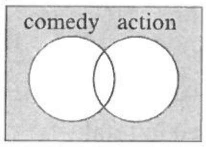



11 What is the probability that a person randomly chosen from the group likes comedy films?

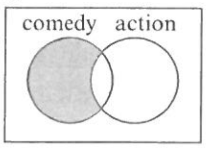
▢ $\frac{18}{25}$ ▢ $\frac{8}{25}$ ▢ $\frac{8}{21}$ ▢ $\frac{18}{21}$

12 Which diagram represents people who do not like action films?

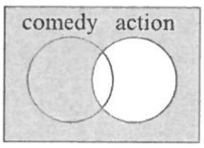
▢ A



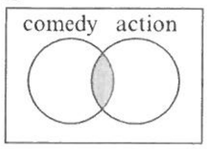

▢ B




▢ C




▢ D



13 Mateo rolled a die 30 times and obtained 11 sixes. He then calculated the experimental probability and made a prediction about what would happen to this experimental probability if he performed 1000 trials. What should his calculation and prediction be?

- experimental probability = $\frac{11}{30}$ which should increase with further trials.
- experimental probability = $\frac{11}{30}$ which should decrease with further trials.
- experimental probability = $\frac{1}{6}$ which should increase with further trials.
- experimental probability = $\frac{1}{6}$ which should decrease with further trials.

14 Two coins are flipped. If this is performed 20 times, what is the expected number of times that both coins would show tails?

15 A new cereal offers a toy surprise with five different toys to find. One toy is twice as common as the other four.
Which table is the best example of potential results of a simulation involving 600 trials?

| Table A | |
|---|---|
| Toy 1 | 100 |
| Toy 2 | 100 |
| Toy 3 | 100 |
| Toy 4 | 100 |
| Toy 5 | 100 |

| Table B | |
|---|---|
| Toy 1 | 120 |
| Toy 2 | 120 |
| Toy 3 | 120 |
| Toy 4 | 120 |
| Toy 5 | 120 |

| Table C | |
|---|---|
| Toy 1 | 214 |
| Toy 2 | 93 |
| Toy 3 | 101 |
| Toy 4 | 106 |
| Toy 5 | 86 |

| Table D | |
|---|---|
| Toy 1 | 113 |
| Toy 2 | 136 |
| Toy 3 | 125 |
| Toy 4 | 119 |
| Toy 5 | 107 |

## ANALYSIS

Eric is researching the popularity of Ford and Holden cars amongst Year 8 boys and girls. He thinks it will be reasonably evenly distributed.

a Draw a tree diagram to represent the situation. (Hint: start the first branches with gender.)

b List the sample space (the possible outcomes) and, assuming all are equally likely, provide the theoretical probability for each outcome.

Eric completes a survey and records these results.

| | Ford | Holden |
|---|---|---|
| Boys | 7 | 23 |
| Girls | 19 | 9 |

c Extend this table to make it a two-way table and find how many people were surveyed in total.

d Find the experimental probability of each outcome.

e Find the probability that a person chosen randomly from the group:

i prefers Holdens to Fords

ii is a boy

iii prefers Holdens to Fords, given they are male.

f Do you think each outcome is equally likely? Explain.

Eric decides to simulate the situation by rolling a die. He assigns each outcome as in this table.

| | |
|---|---|
| Boys who prefer Holden | 1 or 2 |
| Boys who prefer Ford | 3 |
| Girls who prefer Holden | 4 |
| Girls who prefer Ford | 5 or 6 |

g Perform a simulation and comment on how well it represents the experimental data. What limitations exist?

h Challenge! Draw a Venn diagram based on the two-way table.